POWER TORSION DEPENDS ONLY ON THE CLOSED SUBSET: A MODULE IS I-POWER TORSION EXACTLY WHEN ITS SUPPORT LIES IN V(I)

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Editorial note. The proof uses the macros Spec, Ker and the module categories of the shared preamble, all of which the retained preamble supplies verbatim.

Editorial note. The sentence at source line 24106, remarking that the next two lemmas should probably go somewhere else, lies just after the counted proof and is outside the counted spans.

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Allowed standard prerequisites (editorial).

- An element of a module that maps to zero in the localization at f is annihilated by a power of f, and vanishing of the localizations at every prime not containing I is equivalent to vanishing of the localizations at the elements f of I when I is finitely generated. used at proof lines 24095-24100, where the passage from localizations at primes to localizations at the generators f_1, ..., f_r of I is made. Converts the support condition into explicit annihilation by the generators of I.
- The support of an R-module is the set of primes at which its localization is nonzero, and V(I) is the set of primes containing I. used throughout statement and proof, notably at proof line 24097 and at the description of the kernel at proof lines 24101-24102. Provides the vocabulary in which both the hypothesis and the conclusion of the lemma are stated.
- The submodule of elements annihilated by a power of I is the kernel of the map to the product of the localizations at the primes outside V(I).

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used at proof lines 24101-24103. Turns the elementwise equivalence into an identification of the I-power torsion submodule, from which both assertions follow.

Lemma 0.2. *Let R be a ring and let $I \subset R$ be a finitely generated ideal. The subcategory I^∞-torsion $\subset Mod_R$ depends only on the closed subset $Z = V(I) \subset \mathrm{Spec}(R)$. In fact, an R-module M is I-power torsion if and only if its support is contained in Z.*

Proof. Let M be an R-module. Let $x \in M$. If $x \in M[I^\infty]$, then x maps to zero in M_f for all $f \in I$. Hence x maps to zero in $M_\mathfrak{p}$ for all $\mathfrak{p} \not\supset I$. Conversely, if x maps to zero in $M_\mathfrak{p}$ for all $\mathfrak{p} \not\supset I$, then x maps to zero in M_f for all $f \in I$. Hence if $I = (f_1, \ldots, f_r)$, then $f_i^{n_i} x = 0$ for some $n_i \geq 1$. It follows that $x \in M[I^{\sum n_i}]$. Thus $M[I^\infty]$ is the kernel of $M \to \prod_{\mathfrak{p} \notin Z} M_\mathfrak{p}$. The second statement of the lemma follows and it implies the first. $\square$